

Modeling and Global Control of the Longitudinal Dynamics of a Coaxial Convertible Mini-UAV in Hover Mode

J. Escareño · A. Sanchez · O. Garcia · R. Lozano

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Abstract The aim of this paper is to present a configuration for a Convertible Unmanned Aerial Vehicle, which incorporates the advantages of the coaxial rotorcraft for hover flight and the efficiencies of a fixed-wing for forward flight. A detailed dynamical model, including the aerodynamics, is obtained via the Newton-Euler formulation. It is proposed a nonlinear control law that achieves global stability for the longitudinal vertical-mode motion. Indeed, we have performed a simulation study to test the proposed controller in presence of external perturbations, obtaining satisfactory results. We have developed an embedded autopilot to validate the proposed prototype and the control law in hover-mode flight.

Keywords Convertible UAV · Coaxial rotorcraft · Longitudinal control · Global stability · Embedded architecture

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J. Escareño (✉) · A. Sanchez · O. Garcia · R. Lozano
Universite de Technologie de Compiegne,
Centre de Recherches de Royallieu B.P. 20529,
60205 Compiegne, France
e-mail: juan-antonio.escareno@hds.utc.fr

A. Sanchez
e-mail: asanchez@hds.utc.fr

O. Garcia
e-mail: ogarcias@hds.utc.fr

R. Lozano
e-mail: rlozano@hds.utc.fr

1 Introduction

In recent years, the interest on Convertible Unmanned Aerial Vehicles (CUAVs) has increased since the operational scope is wider than that provided by conventional designs. These vehicles combine the maneuverability of the helicopter to deal with confined spaces (hover, vertical take-off and landing) and the forward-flight advantages of the airplane (speed and range). The flight profile of CUAVs improves the performance of missions like: the search and rescue of people in hazardous locations or circumstances (earthquakes, spills and fires); and surveillance or information collection on important installations in either highly sensitive locations (borders, ports and power-plants) or remote or uninhabitable places (polar zones, deserts and toxic areas). A logical drawback arising from single-propeller aircrafts is the vulnerability to reacting torque; to counteract this effect a complementary input of angular momentum is needed. Therefore, helicopters vehicles waste some power to feed a tail-rotor, while fixed-wing aircrafts, in hover flight, use differential control surfaces (elevon or ailerons). An alternative way to face the reacting torque, without using complementary controls, is the use of counter-rotating propellers. This configuration increases the thrust and reduces the size of the vehicle but it increases the mechanical complexity. In the CUAV's field, the tailsitter configuration stands out among other CUAVs configurations (tiltrotor, tiltwing, etc.) because it lacks of swashplate or tilting elements, instead it uses the propeller to wash the normal aircraft surfaces (elevator, aileron and rudder) so that the aircraft can generate the torques to handle vertical flight. However, in vertical-flight regime, this configuration is very vulnerable to wind gusts, which reveals the importance of robust hover flight for this kind of vehicles. We have fused the previous concepts to obtain a vehicle that offers the mechanical simplicity of tail-sitter and the operational flexibility of coaxial rotorcrafts. The topic of CUAVs is currently addressed by numerous research

Fig. 1 Experimental convertible coaxial UAV: Twister

